

A Live-Coded Add-On System for Video Conferencing in Virtual Reality

Septian Razi*

Australian National University

Henry Gardner†

Australian National University

Andrew Sorensen‡

MOSO Corporation

Matt Adcock§

CSIRO's Data61

ABSTRACT

Despite our increasing reliance on video conferencing, it has inherent limitations in engagement and effective communication. We explore a VR add-on system that supplements traditional video conferencing where a host live-codes data visualisations for stakeholders. It combines VR visualisation techniques with libraries present in popular data analytics tools such as Python, allowing real-time changes by utilising a RESTful realtime database. An application has been built to explore pedigree node-link graphs, and its feasibility has been analysed with a few domain experts. Trials have demonstrated that our system appears to enhance engagement and communication above traditional video conferencing for data exploration.

Index Terms:

Human-centered computing—Visualization—Visualization techniques—Graph drawings; Human-centered computing—Visualization—Visualization application domains—Scientific visualization; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality.

1 INTRODUCTION AND BACKGROUND

In current times people have become increasingly reliant on video conferences as a means of communication. However, this format has inherent limitations because of its relative lack of engagement and presence. Because Virtual Reality addresses human engagement and sense of presence at its core, there are developments underway to incorporate modern expectations for video conferencing into VR. For example, Facebook's Horizons Workrooms [4] combines a 3D virtual meeting world with streaming video tiles, live presentations and a collaborative whiteboard.

Live coding refers to systems that enable a programmer to apply programmatic changes to a computer program and to see the results of those changes in perceptually-real time. The sensed immediacy and power of such systems increase engagement, and are regularly used in education [8]. Live coding of VR environments has also been demonstrated in the literature, particularly for creative, design and performative applications [2, 6].

In this paper we describe our experience of building a VR add-on to a video conferencing scenario where a host live codes visualisations of data for participants.

2 SYSTEM DESCRIPTION

Our system operates as an add-on to a traditional video conferencing tool and consists of three main components, the Desktop Interface, a Realtime Database and a Virtual Reality Application. This system as whole allows users to transfer data between these components as an aide to data exploration. We allow the video conference host to operate the desktop interface whilst concurrently communicating with the participants. The host is able to live-code and send data

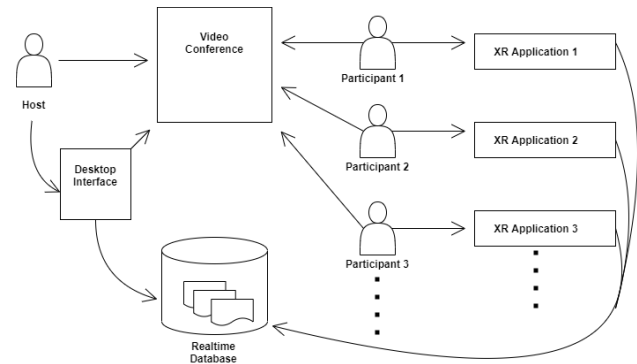


Figure 1: Schematic view of the system

via the real-time database from the notebook to the participants possessing a VR HMD and explore the data themselves within a VR application.

2.1 Components

2.1.1 Desktop Interface

Our Desktop interface consists of a small utility libraries to be utilised in Python 3 Notebooks commonly used by scientists and data engineers. These utilities allow users to translate the data created by the desktop programs into a format that can be transferred through to the Realtime Database. Our desktop interface utilises iGraph [5] to encode data into a graph format and utilises the visualisation algorithms present in matplotlib [3], this data of which is transferred to the other interfaces via the inbuilt RESTful HTTPS requests to the Realtime Database.

2.1.2 VR Application

The 3D layout of the graph is controlled by the Desktop interface, via data transfer through the Realtime Database. The VR application interprets this data and visualises it in a manner that is mapped to the visualisation on the desktop interface, similar to those described in [1, 7]. Our VR application also incorporates interaction techniques from highlighting, rotation, scaling and other useful techniques available within a VR system. We also include a history of visualisations retrieved from the desktop interface. This ability to interact and change the graph visualisations gives users agency in further exploring the data and allows users to be able to compare visualisations and refer back to previous ones.

2.1.3 Realtime Database

Our realtime database serves as the main medium to communicate between our Desktop Interface and Virtual Reality Application. The realtime database utilises RESTful API calls to facilitate data transfer. Our data consists of JSON objects representing both the graph and its layout in 3D. We utilised RESTful APIs as the main communication method between the desktop interface and the VR application.

*e-mail: septian.razi@anu.edu.au; ORCID 0000-0003-2122-7362;

†e-mail: henry.gardner@anu.edu.au; ORCID 0000-0002-8613-8777;

‡e-mail: andrew@moso.com.au

§e-mail: matt.adcock@csiro.au; ORCID 0000-0001-6191-9887;

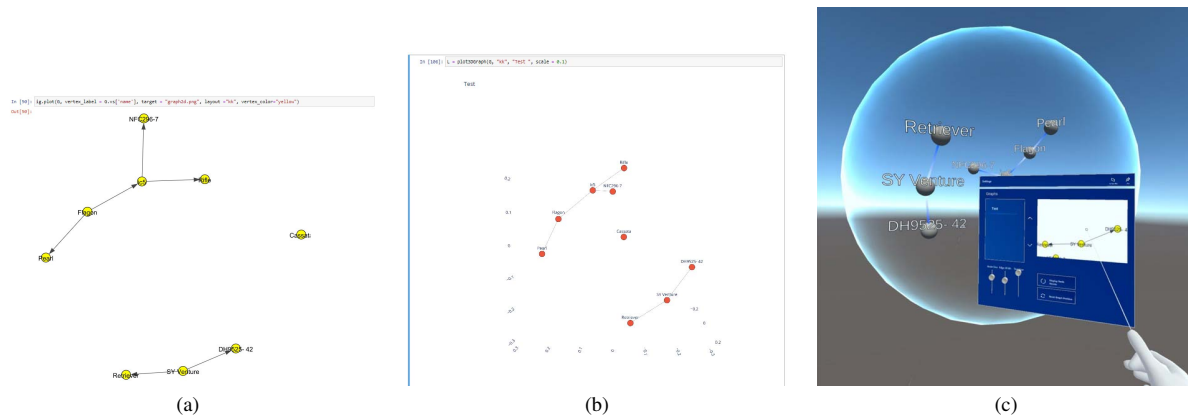


Figure 2: Example of System Data Transfer (a) Showcases the 2D image generated in the Desktop Interface to be passed onto the VR application (b) 3D visualisation on Python Notebook of the same graph (c) Resulting Data in the VR application, with the 3D layout retrieved from (b) and the image seen on the panel originating from (a)

3 USER STUDY

3.1 Configuration

We conducted three distinct initial studies with one unique participant each. Our participants consisted of domain experts within the field of bioinformatics, one of whom is an expert at plant pedigree visualisation. One of the studies of the participants utilised an Oculus Quest in combination to Cisco Webex, and two others utilised an Oculus Quest 2 with Zoom Conferencing. All were full-time scientists working in a field related to bioinformatics (1 female, 2 male). Each had differing levels of expertise with the dataset we used.

The dataset we utilised originates from the sample datasets provided for the Helium Project [9]. The two dataset we used were the Golden Promise 2019 dataset and the UK Barley Recommended List Pedigree. The Golden Promise 2019 Dataset is a smaller dataset consisting of 43 nodes and 45 edges, whereas the UK Barley Recommended List dataset is larger with 83 nodes and 78 edges.

3.2 Experiment

For each session, we had one (host) participant controlling the Desktop Interface, another (audience) participant had access to the VR Application and one additional (audience) participant as an observer. Our emphasis within this user study was to explore the feasibility of our system by analysing the quality of the interaction and communication. We implemented a qualitative approach where all sessions were recorded and transcribed and notes were taken by the host and observer.

We had an overall structure with some possible tasks relating to exploring the graph dataset, however we also prioritised more natural exploration from the participants. The tasks included: Finding Neighbours of a given node; Finding a Path between two nodes; Finding the node with the most neighbours; Finding how many distinct “clusters” exist in the node; Recalling node spatial location and were derived from [7] as a means to measure understanding and engagement for graph data.

After the experiment, we sent participants a post-study questionnaire which requested Likert-scaled ratings of statements pertaining to engagement, communication, data understanding, comfort and perceived usefulness of the VR application.

4 RESULTS AND CONCLUSION

In our user study, our system showed promising results in terms of engagement. All of our expert participants gave engagement the highest ranking of 5 and this was supported by interview transcripts.

Communication was also shown to be enhanced. Qualitative analysis of interview transcripts and the corresponding questions in the post-test survey showed that communication regarding the data was able to be maintained, and if not enhanced, when participants went into VR.

Despite its promise, there are still some pain points and room for improvement in our system. One participant remarked that they were frustrated with the way that the graphics on the desktop looked but, in our system, they had no means to directly affect the desktop as that was under the control of the conference host who was not able to directly view the issue. Overall, it would be useful for participants to send data and information back to the host session.

Our research is promising for the future of immersive analytics and VR video conferencing systems. While there is a big commercial push for complete VR conferencing systems where users would be perpetually in a virtual world, our system shows that an engaging VR conferencing experience does not necessarily require such a big step to maintain accessibility. Similar considerations of our study would also be relevant for applications that have online education as their objective or other application areas.

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